The Crowded Room

If the idea that many users can coexist in the same spectrum sounds counter-intuitive, another analogy should help. Wireless communication in the radio-frequency spectrum is fundamentally similar to wireless communication in the acoustic spectrum, otherwise known as speech.

Imagine a group of people in a room. Experience tells us that everyone can carry on a conversation with his or her neighbor simultaneously, even with music playing the background, so long as people speak at a normal volume. If someone starts yelling, he or she will drown out other speakers, who will be forced to speak louder themselves in order to be heard. Eventually, some portion of the room simply won't be able to communicate over the background noise, and each additional person who starts yelling will reduce the total number of conversations.

We could call that a "tragedy of the commons." We could enact laws giving only some individuals the right to speak during defined times, ensuring they can shout as loud as they want without interference. But that would clearly be an unnecessary solution with significant negative consequences.

Think back to the situation where everyone is speaking in a normal tone of voice. What allows so many conversations to occur simultaneously is that the people talking are modulating their communications in an appropriate way, and the people listening are able to distinguish one conversation from another. It's the intelligence at the ends of the conversation, not the integrity of the signal, that allows for more efficient communication. The same is true in the radio frequencies. Intelligent devices can distinguish among more simultaneous transmissions than simple ones. The more sophisticated and agile the system, the more the overall carrying capacity of the spectrum increases.

One might protest that the speaking analogy breaks down